

# **Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based**

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## METHODOLOGY

### 1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

| Parameter          | Sentinel-1 C-SAR (IW Mode)      |
|--------------------|---------------------------------|
| Frequency Band     | C-band (~5.405 GHz)             |
| Revisit Time       | 12 days (single satellite)      |
| Spatial Resolution | 5 m (range) × 20 m (azimuth)    |
| Coverage Mode      | Interferometric Wide Swath (IW) |

### 2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

|                                                                     |
|---------------------------------------------------------------------|
| S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D |
| S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431 |
| S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6 |
| S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312 |
| S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14 |
| S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F |
| S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445 |
| S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52 |
| S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD |
| S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8 |
| S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13 |
| S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2 |
| S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5 |
| S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D |
| S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD |

### 3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



## VISUALIZATIONS

### 1. Defining specific bridge location & parameters for satellite SAR based inspection



#### Project & Bridge Details

State – Rajasthan  
Bridge No. – 895/2  
Bridge Chainage – 924+690

Project - Chittorgarh to Kota  
Zone – RJ 06  
Structure Type – MNB

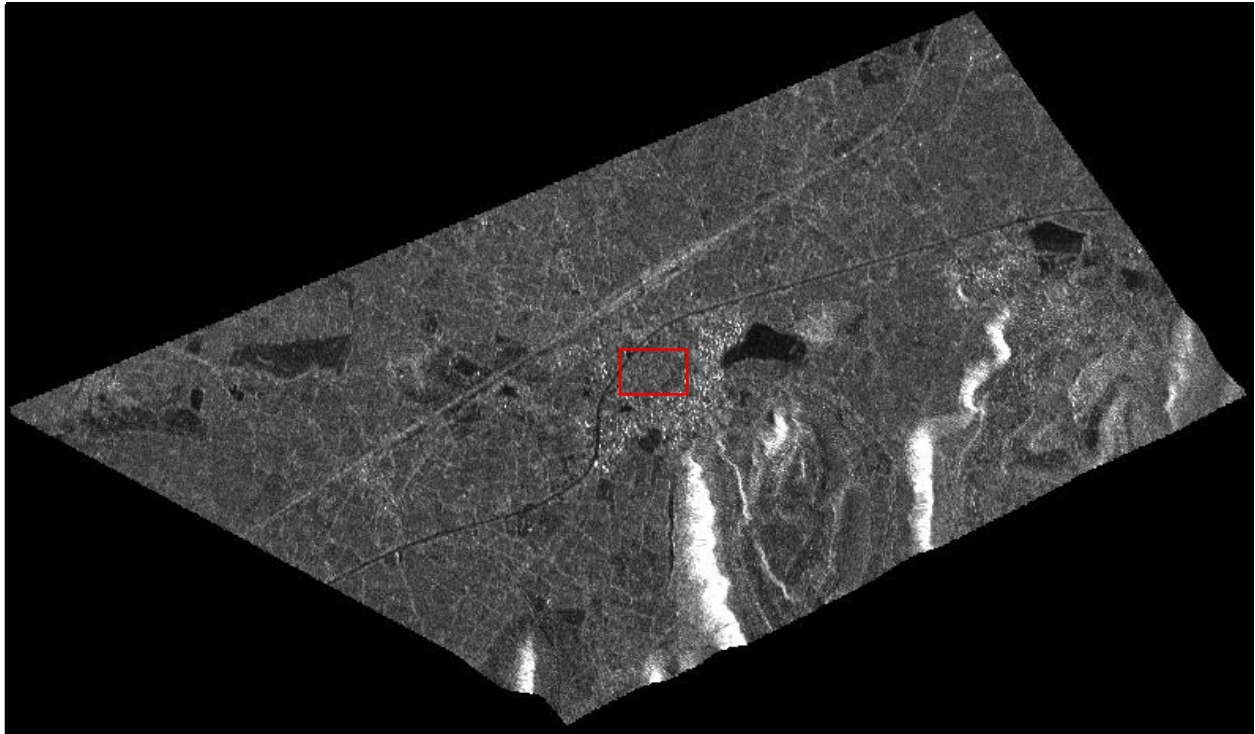
#### Sar Parameters

Sensor – Sentinel-1 (C-Band)  
Beam Mode – IW (Interferometric Wide Swath)  
Path – 34  
Frame – 519

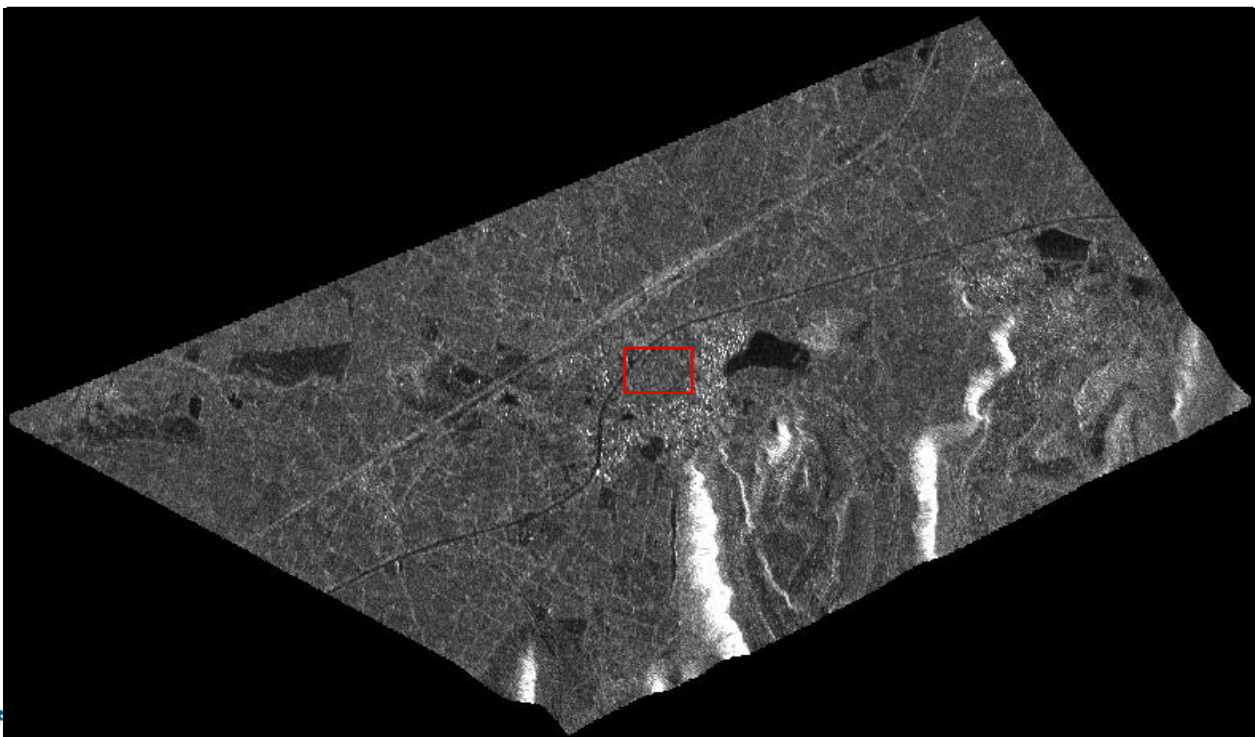
Flight Direction - DESCENDING  
Polarization – VV+VH  
Absolute Orbit – 54706

## 2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

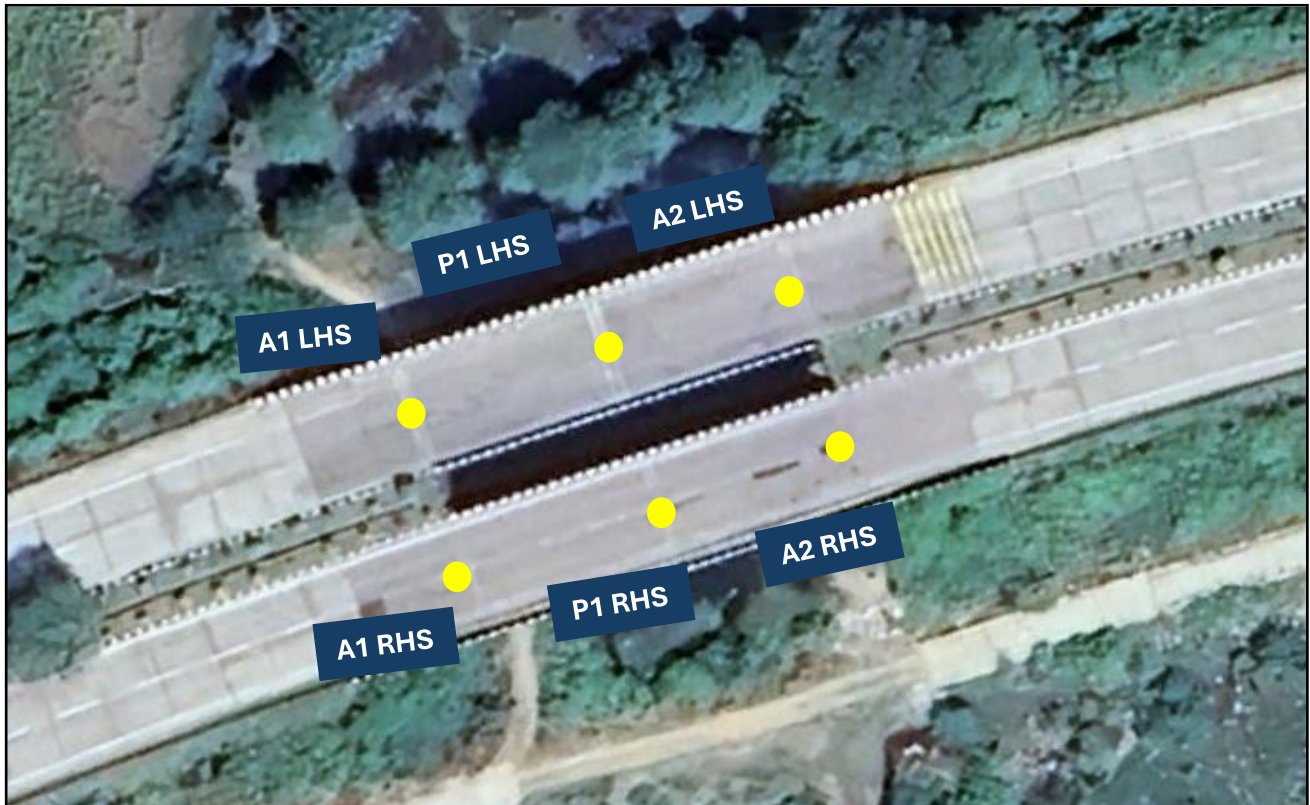


### 3. Results of SAR based bridge inspection, including visualizations

#### Bridge Components

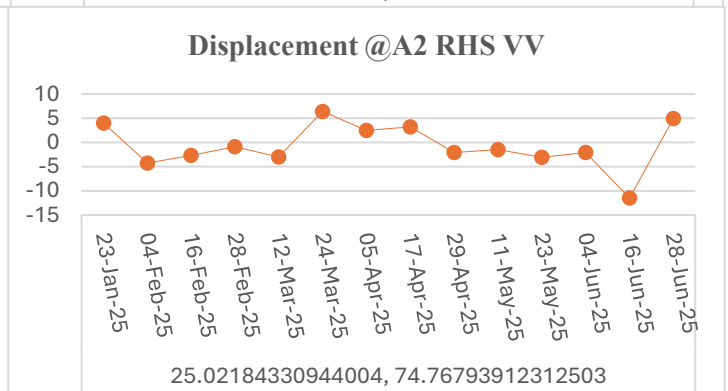
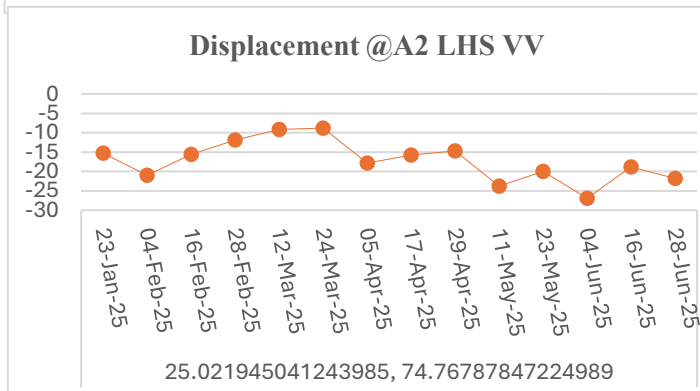
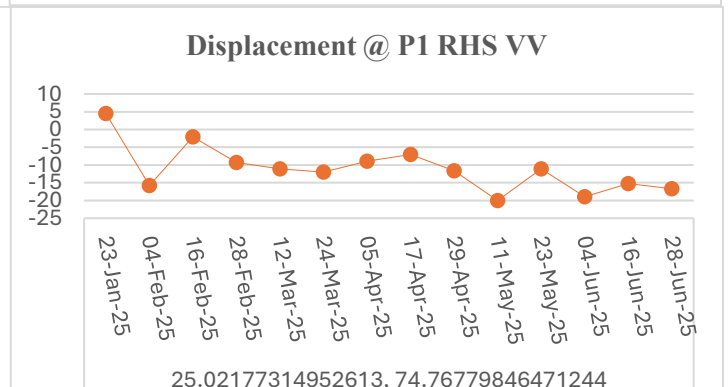
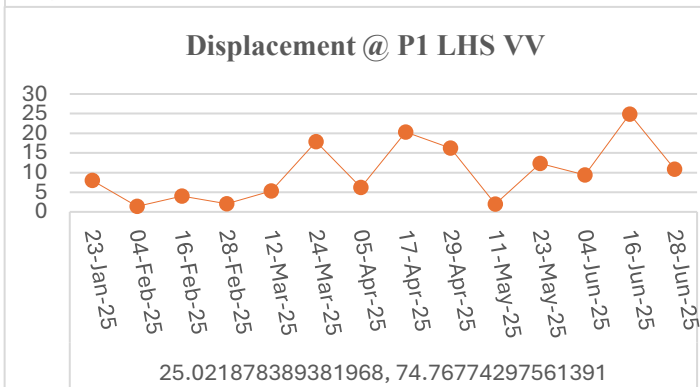
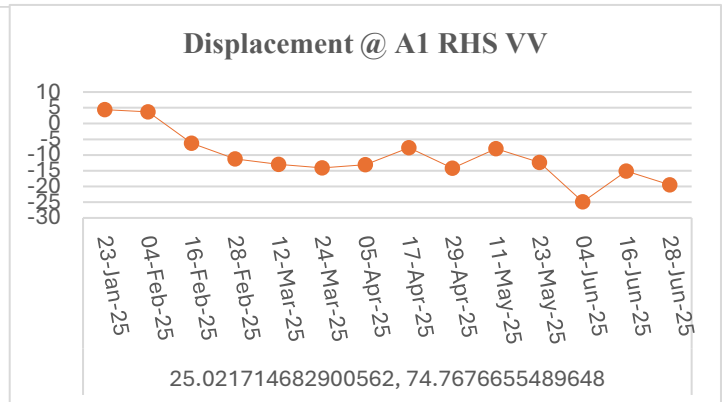
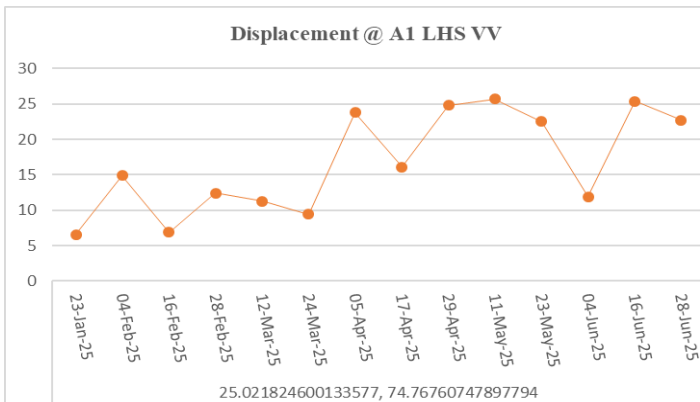
MNB – CH-924+690

Length – 55.30M

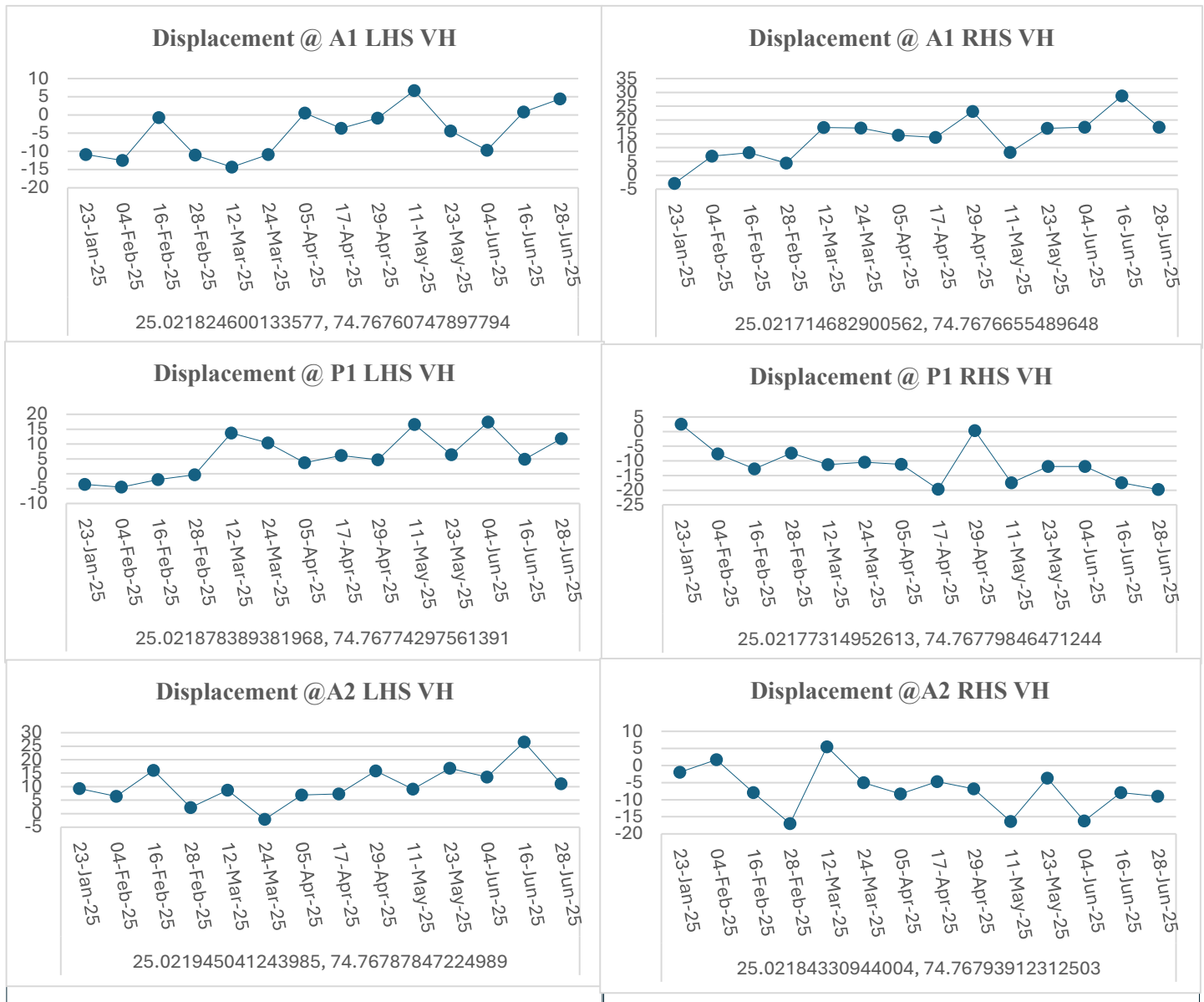


|                           |  |                    |                   |
|---------------------------|--|--------------------|-------------------|
| Scan Duration             |  | From : 23 Jan 2025 | To : 28 June 2025 |
| Components & Abbreviation |  |                    |                   |
| A1                        |  | Abutment 1         |                   |
| P1                        |  | Pier 1             |                   |
| A2                        |  | Abutment 2         |                   |
| LHS                       |  | Left Hand Side     |                   |
| RHS                       |  | Right Hand Side    |                   |

### 3.1 VV Displacement From 23 Jan 2025 to 28 June 2025



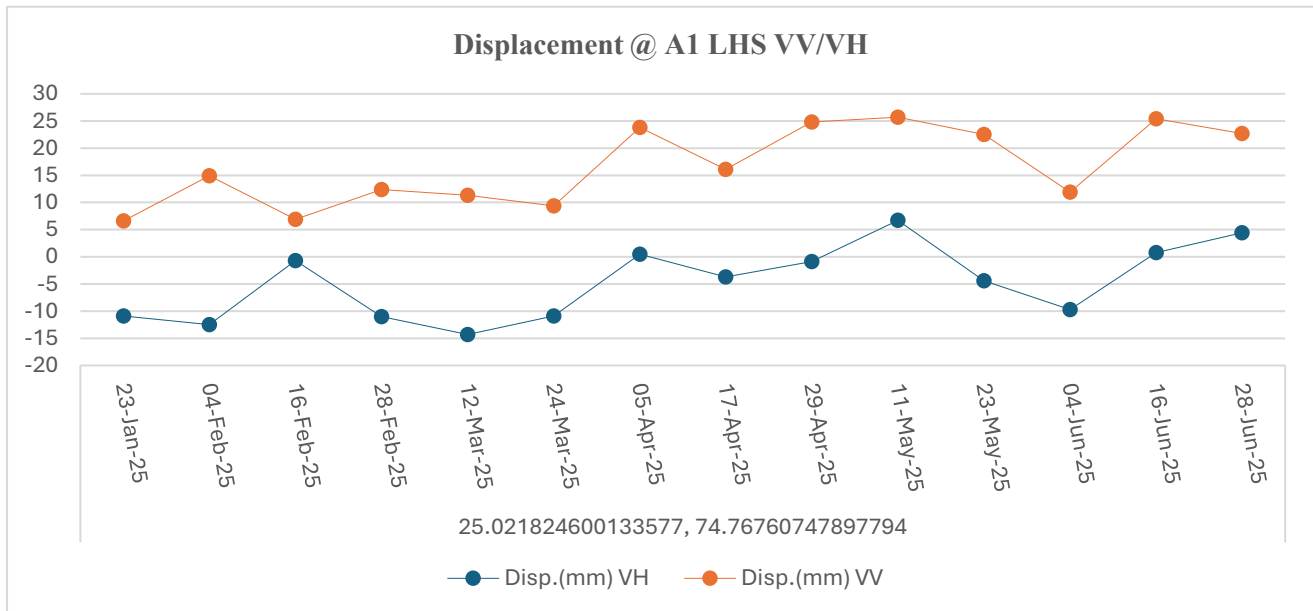
### 3.2 VH Displacement From 23 Jan 2025 to 28 June 2025



### 3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.021824600133577, 74.76760747897794)



#### Key Observations:

##### Displacement Magnitude Patterns:

- Vertical Displacement (VV)** shows a consistently positive trend, generally ranging from 7 mm to 26 mm. The VV series demonstrates repeated uplift cycles with several pronounced peaks, indicating sustained vertical movement dominance throughout the monitoring period.
- Horizontal displacement (VH)** exhibits a predominantly negative to mildly positive oscillatory behaviour, constrained roughly between -15 mm and +7 mm.

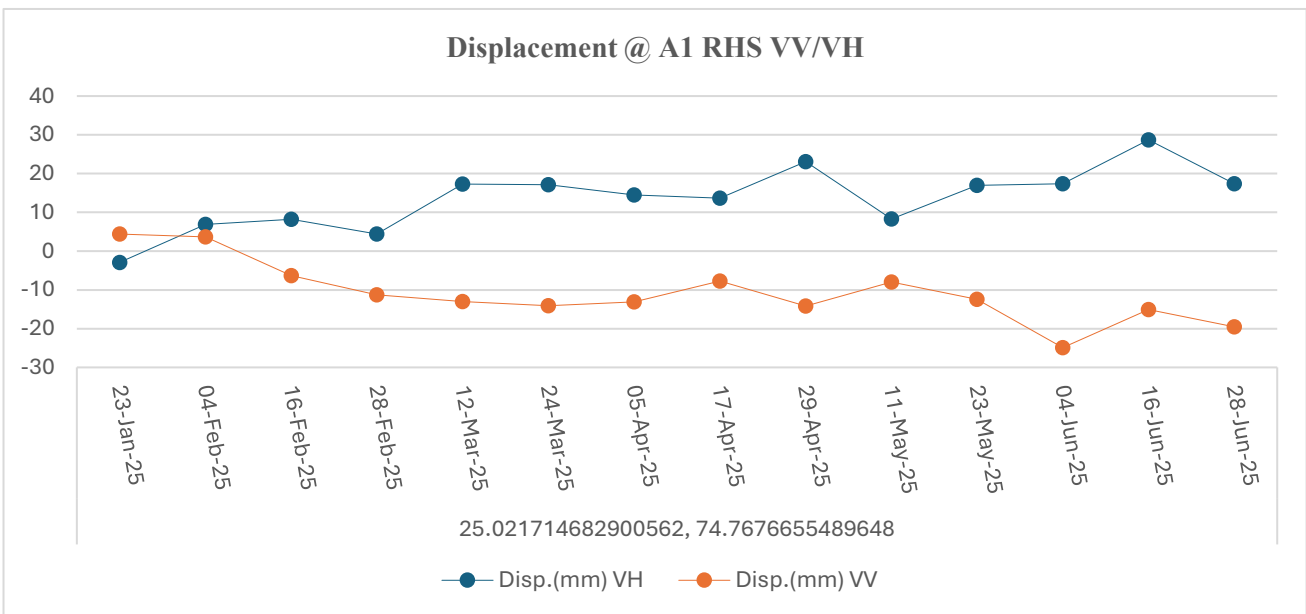
##### Critical Events & Anomalies:

- February 2025:** VH records a pronounced negative excursion (-15 mm), marking the most significant horizontal displacement event in the dataset potentially indicative of lateral stress release, ground relaxation.
- June 2025:** Renewed divergence is observed, with VV maintaining elevated uplift (22–25 mm) while VH returns to moderate positive values (4–6 mm), suggesting vertical dominance with reduced horizontal restraint.

### 3.4 SAR Based Analysis at Abutment 1 RHS:



**A1 RHS** (25.021714682900562, 74.7676655489648)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** demonstrates a predominantly positive and increasing trend, ranging approximately from -5 mm to +28 mm.
- **Vertical displacement (VV)** is consistently negative throughout the monitoring period, spanning roughly -5 mm to -25 mm, indicating ongoing settlement or downward movement with moderate variability.

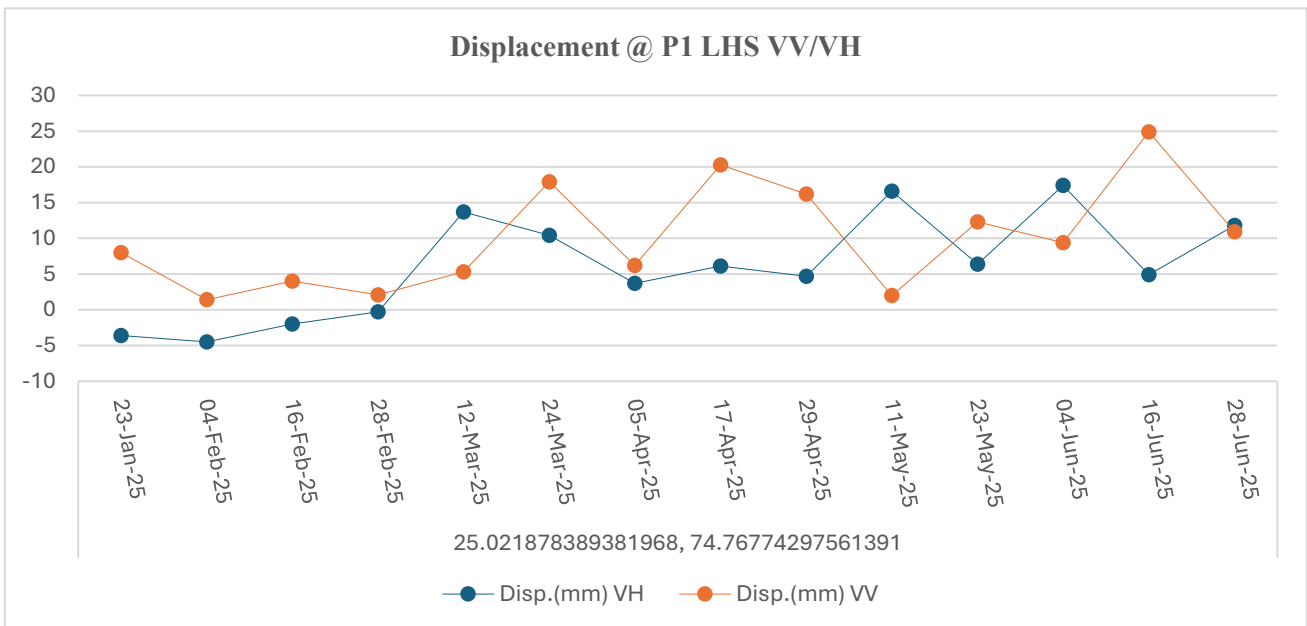
##### Critical Events & Anomalies:

- **February to Mid-March 2025:** VH exhibits a rapid increase from 5 mm to 18 mm, suggesting an acceleration in horizontal structural response or ground deformation.
- **June 2025:** Both VH and VV show partial moderation, suggesting a temporary stabilization or redistribution of stresses.

### 3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (25.021878389381968, 74.76774297561391)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows a moderate oscillatory behaviour, varying roughly between -5 mm and +18 mm. VH movements are generally positive after late February, suggesting gradual lateral mobilization with intermittent reversals.
- **Vertical displacement (VV)** displays a highly variable but predominantly positive pattern, ranging approximately from +2 mm to +25 mm.

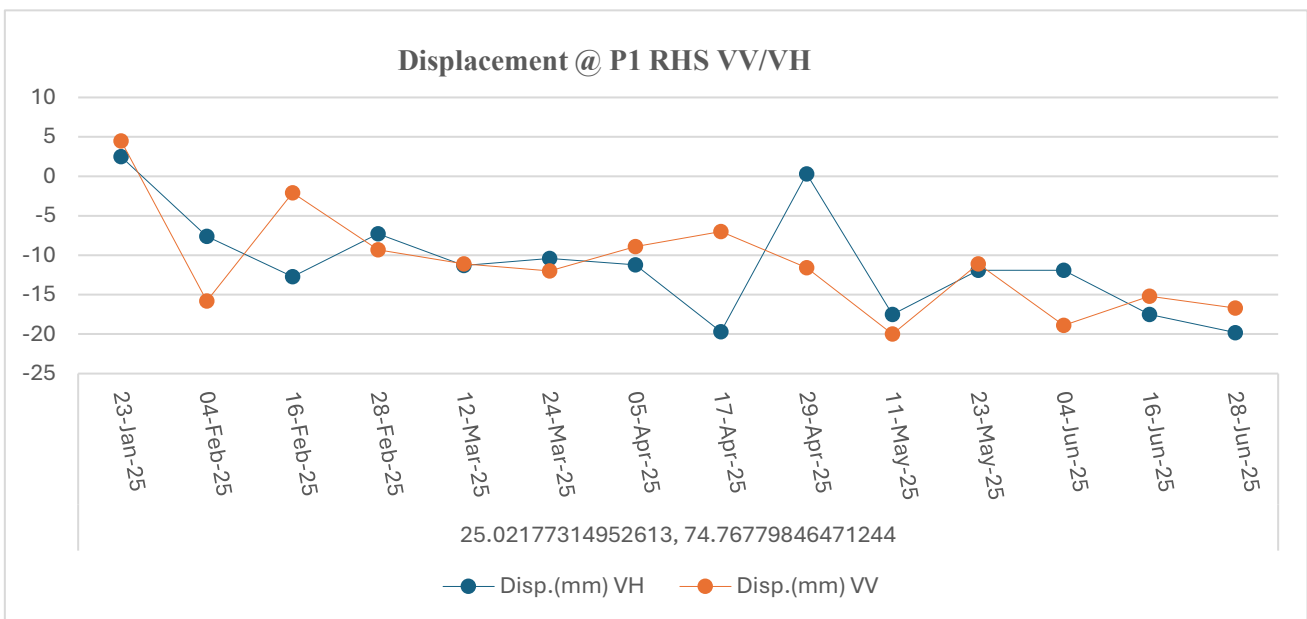
##### Critical Events & Anomalies:

- **February to Mid-March 2025:** Both VH and VV rise sharply, indicating a coupled displacement response likely driven by a common loading or ground response mechanism.
- **June 2025:** Both VH and VV converge toward moderate positive values (10–12 mm), suggesting a stabilization phase following earlier peaks.

### 3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (25.02177314952613, 74.76779846471244)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** The horizontal displacement (blue line) fluctuates between approximately +5mm and -5mm, with a general trend of fluctuation around the zero value.
- **Vertical displacement (VV)** The vertical displacement (orange line) shows more significant variation, fluctuating from +5mm to -20mm.

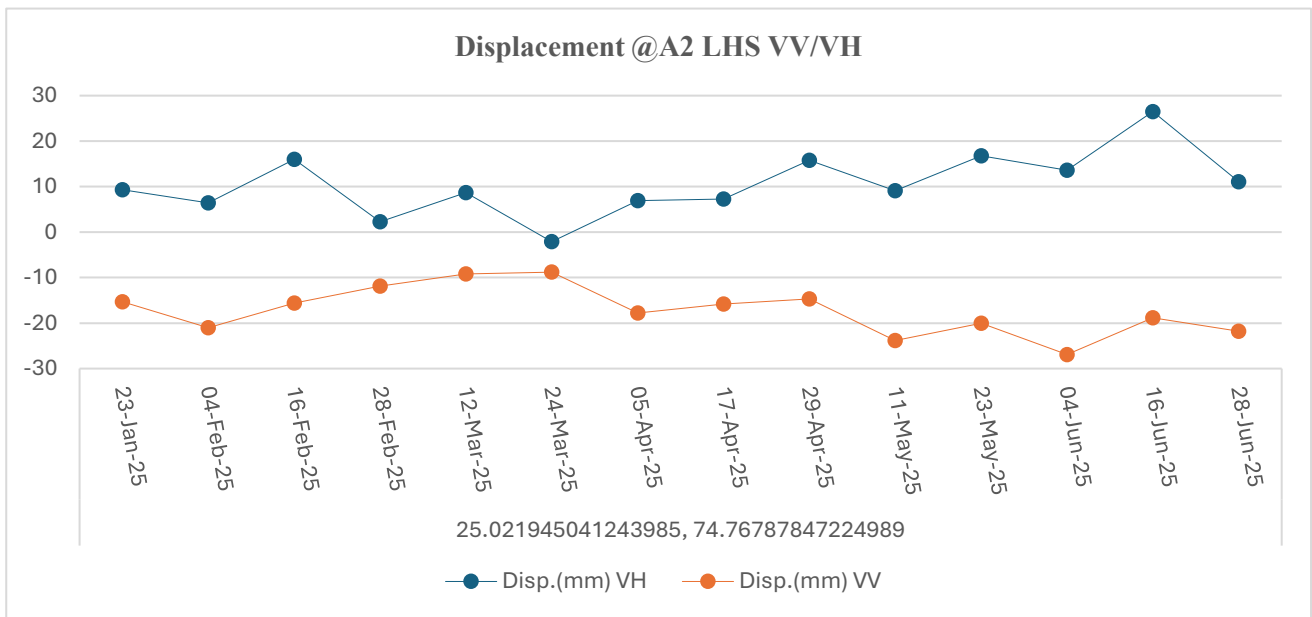
##### Critical Events & Anomalies:

- **Significant downward shift:** In the middle of the observation period (around March 2025), a substantial downward shift is observed for the vertical displacement, with a dip reaching near -20mm. This indicates a significant downward structural movement that requires further investigation.
- **Moderate Stabilization:** in the later periods (April to June 2025), both displacement components (VH and VV) show signs of stabilization, with the VH moving within a narrow range between +2mm and +5mm, while the VV remains relatively stable near the -5mm mark.

### 3.7 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.021945041243985, 74.76787847224989)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** The horizontal displacement (blue line) varies between approximately +5mm and +15mm, showing a relatively stable trend with minor fluctuations.
- **Vertical displacement (VV)** The vertical displacement (orange line) shows considerable fluctuation, ranging from about +5mm to -30mm.

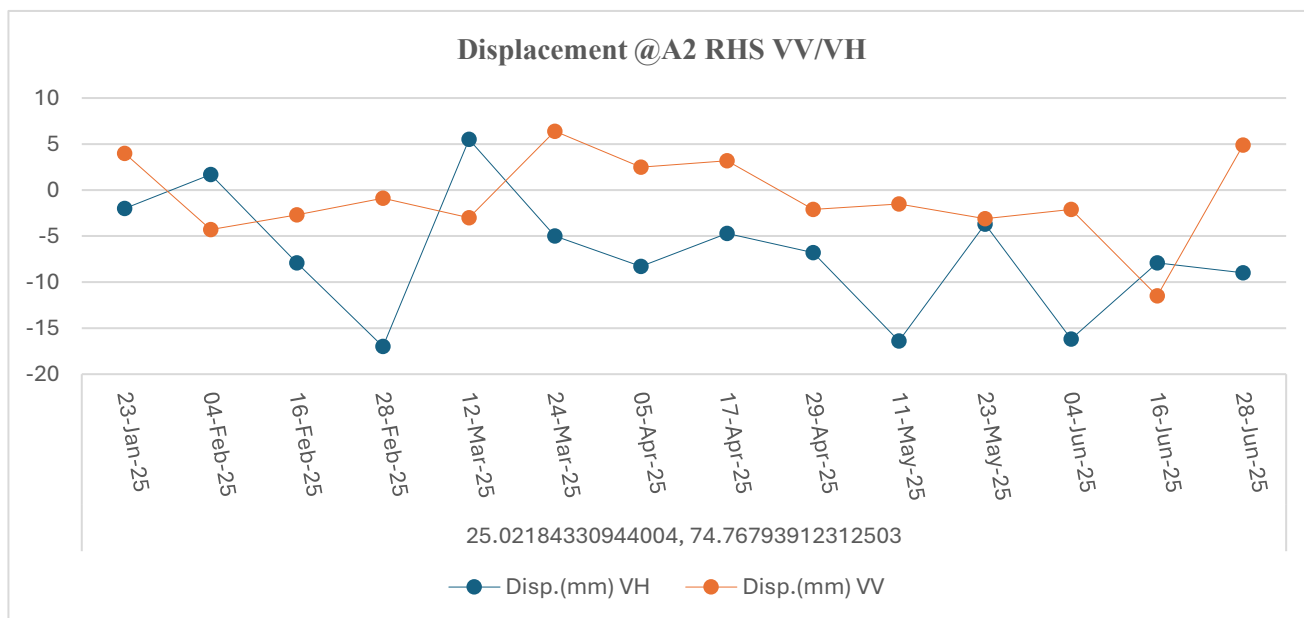
##### Critical Events & Anomalies:

- **Significant downward shift:** There are several noteworthy downward spikes in the vertical displacement, especially around late February (approaching -30mm) and mid-April (around -25mm to -30mm).
- **Moderate upward shift in VH:** Horizontal displacement (VH) shows a noticeable upward movement around mid-June 2025, where it peaks at around 15mm. This is an anomaly that should be monitored to ensure no unexpected shifts are occurring in the structure.

### 3.8 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.02184330944004, 74.76793912312503)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** values fluctuate between approximately +5 mm and -18 mm, indicating moderate variability with a generally negative bias.
- **Vertical displacement (VV)** displacement ranges from about -12 mm to +6 mm, showing lower volatility compared to VH.

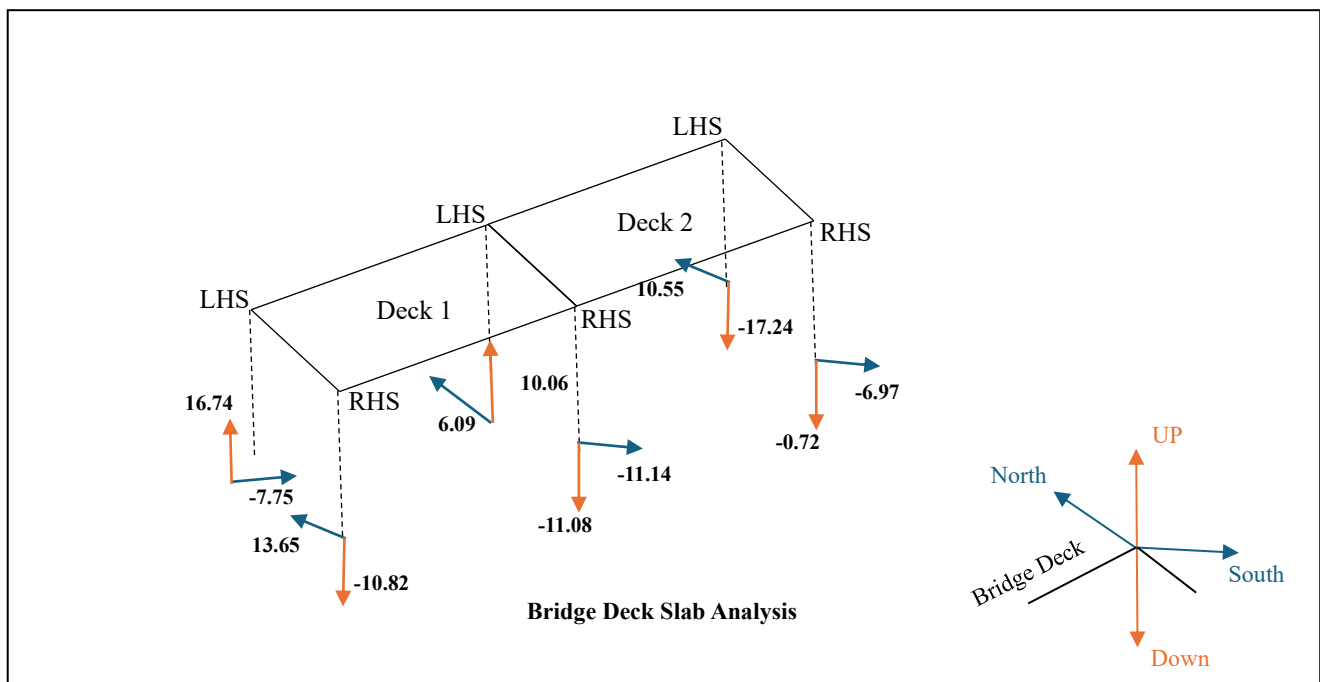
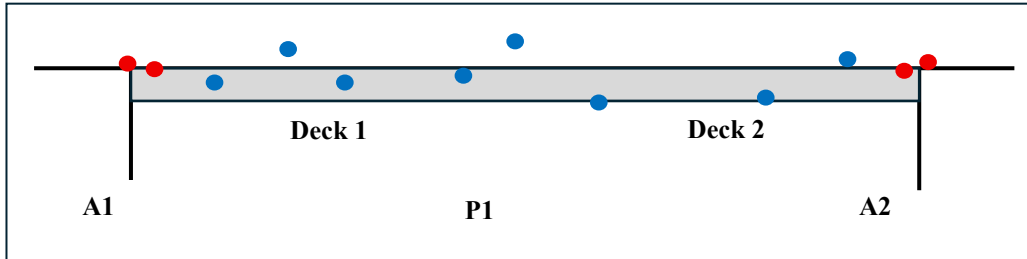
##### Critical Events & Anomalies:

- **Major lateral movement events:** Pronounced negative VH spikes around late February (-17 mm) and mid-May (-16 mm) indicate significant horizontal shifts that may reflect localized structural response or external loading.
- **Vertical Recovery Phase:** A noticeable positive VV rebound in late June (+5 mm) suggests partial vertical recovery or uplift following earlier minor settlement.
- **Recent Behaviour:** In the final monitoring periods, VH remains consistently negative while VV trends upward, indicating horizontal movement dominance with relatively stable vertical response.

## ANALYSIS

### 1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



#### Vertical Displacement Behaviour:

- The deck slab exhibits noticeable vertical displacement variability across Deck 1 and Deck 2, with both uplift and settlement observed at LHS and RHS locations.
- The alternating uplift and settlement patterns between adjacent supports suggest localized differential behaviour, potentially influenced by bearing performance, load distribution.

#### Horizontal Displacement Behaviour:

- Lateral (north–south) displacements vary significantly between deck segments, with magnitudes ranging from approximately –13.65 mm to +10.55 mm. Deck 1 shows a combination of opposing lateral movements between LHS and RHS, indicating torsional response or asymmetric loading. Deck 2 displays comparatively consistent lateral trends, though measurable drift remains present, suggesting ongoing accommodation of thermal, creep, or traffic-induced effects.

**CONCLUSION:** The deck slab displacement patterns demonstrate meaningful structural movement, with vertical and horizontal responses exceeding what would typically be attributed solely to uniform thermal expansion. Differential behaviour between Deck 1 and Deck 2, as well as between LHS and RHS supports, highlights the influence of asymmetric loading and varying boundary conditions.